

ECE264D:

CMOS Analog Integrated Circuits and Systems IV

Lecture 3:

Analog PLL – Phase Frequency Detector, Charge Pump and Type-II PLL

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Outline

- Why Type-II PLL?
- Phase frequency detector (PFD)
 - Functionality and implementation
 - PFD and PD comparison
- Charge pump
 - Functionality and implementation
 - Nonidealities
- Type-II PLL
 - S-domain model and loop transfer functions
 - Loop stability
 - Loop dynamics



Issues with Type-I PLL

- Limited acquisition range
 - When initial frequency difference between VCO and reference is large, PD output is periodic and has an average close to zero. The loop may never acquire lock.
 - PD gain is negative for part of input range that makes the loop positive feedback
- The loop stability (damping factor ζ and phase margin PM) is closely tied to the loop gain and loop bandwidth.
 - Lead to tradeoff between stability and loop bandwidth
 - Loop bandwidth needs to be independently optimized for phase noise, reference spurs, etc.
- Finite static phase error between reference and VCO that is VCO frequency dependent

$$\Delta\phi = \frac{\omega_{VCO} - \omega_0}{K_{PD} \cdot K_V}, \text{ where } \omega_{VCO} = \omega_0 + K_V \cdot V_{tune}$$

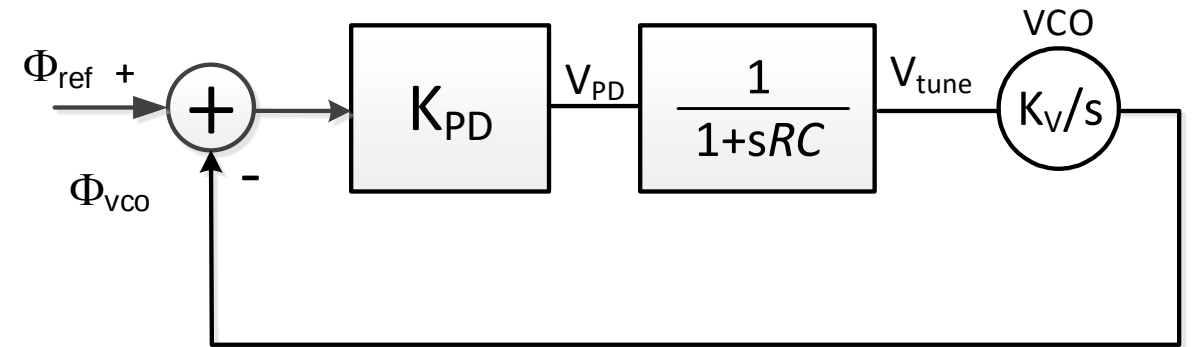
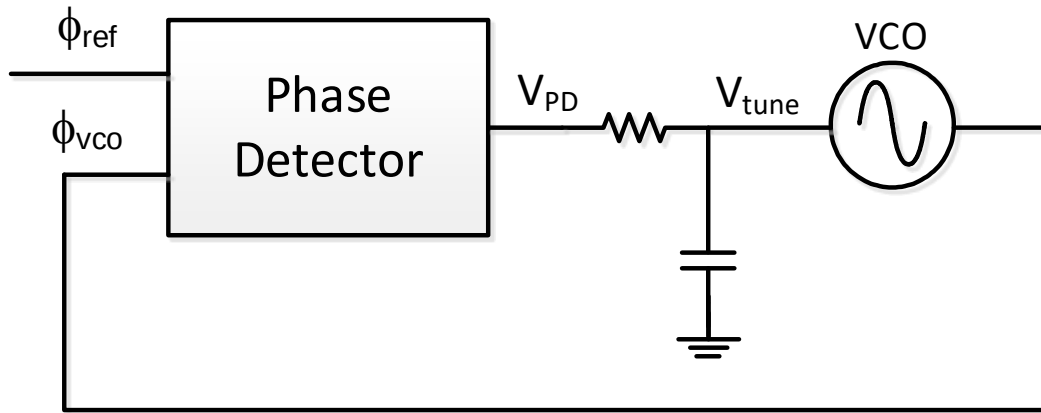


Solutions by Type-II PLL

- Phase frequency detector (PFD) instead of phase detector (PD)
 - Detect both phase and frequency
 - No PFD gain polarity ambiguity that leads to positive feedback loop
 - Detection based on input signal rising edge only: insensitive to duty cycle
- Decouple loop phase margin and loop bandwidth
- Two poles at DC in open loop transfer function
 - In lock state, both phase and frequency errors are zero



Revisit Type-I Phase Locked Loop S-domain Model



- S-domain model of open loop transfer function

$$H_{OL}(s) = \frac{K_{PD} \cdot K_V}{s(1 + sRC)}$$

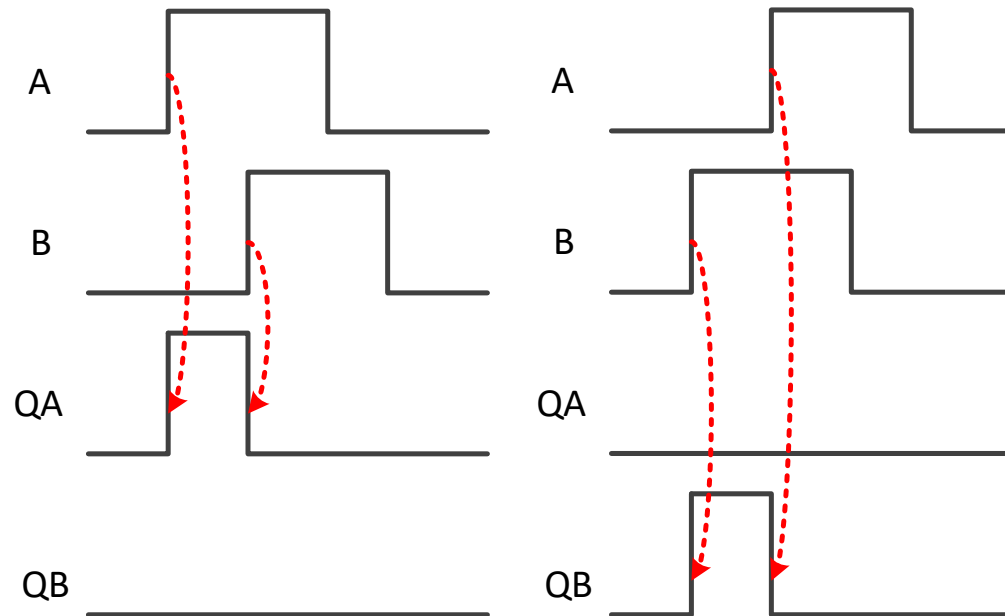
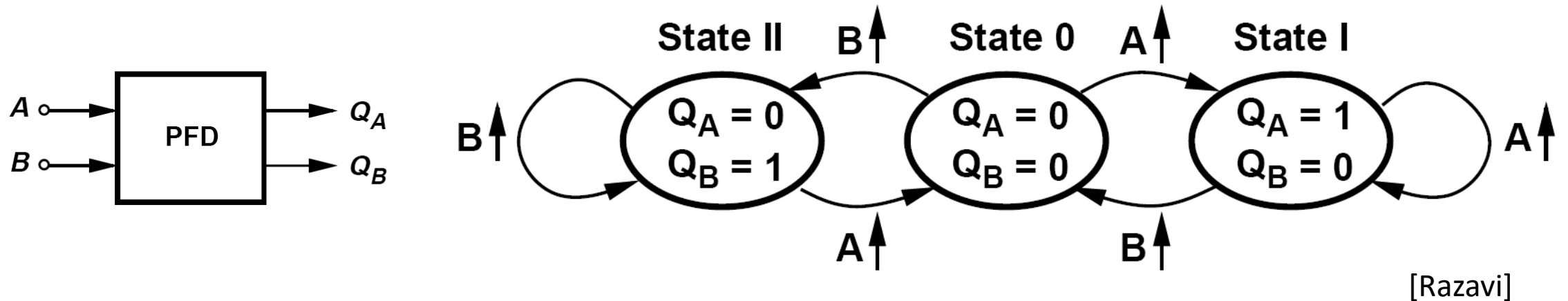
- Closed loop transfer function

$$H_{CL}(s) = \frac{\Phi_{VCO}}{\Phi_{ref}} = \frac{H_{OL}}{1 + H_{OL}} = \frac{\frac{K_{PD} \cdot K_V}{RC}}{s^2 + \frac{s}{RC} + \frac{K_{PD} \cdot K_V}{RC}}$$

- One pole at DC in open loop transfer function: **Type-I PLL**



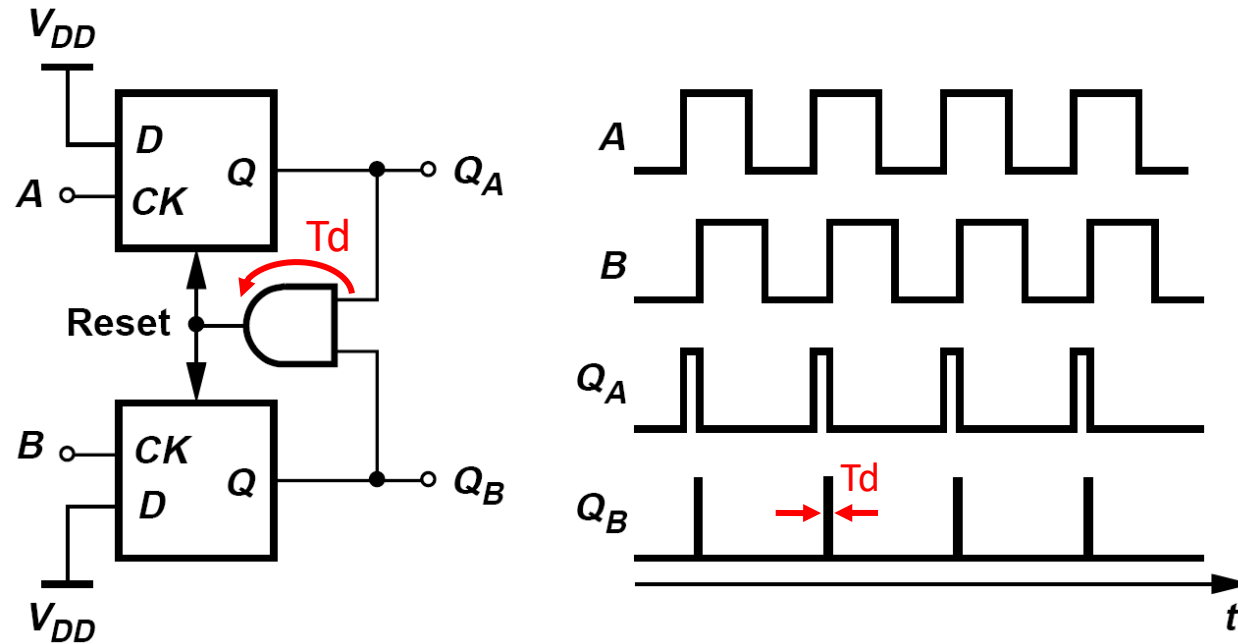
Phase and Frequency Detector



- A rising edge on A generates
 - A rising edge on Q_A if Q_A is low
 - A falling edge on Q_B if Q_B is high
- A rising edge on B generates
 - A rising edge on Q_B if Q_B is low
 - A falling edge on Q_A if Q_A is high
- The average of " $Q_A - Q_B$ " represents the phase and frequency difference between A and B



PFD Implementation

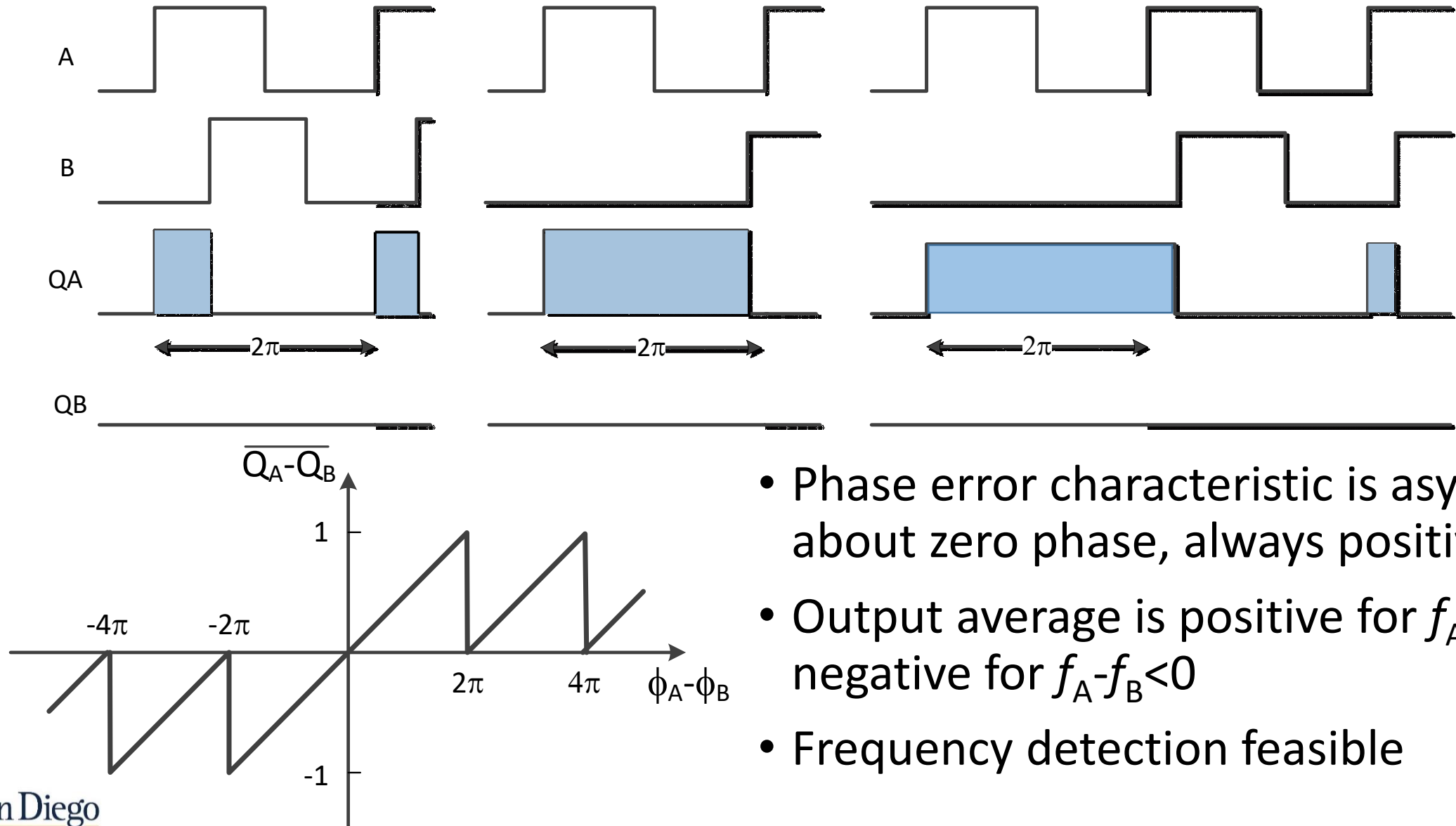


[Razavi]

- PFD consists of two D-flip-flops (DFF's) and an AND gate
- A and B are clock signals; D inputs are logic ONE.
- A rising edge on A latches Q_A to D (logic ONE)
- A rising edge on B latches Q_B to D (logic ONE)
- When both Q_A and Q_B are high, AND gate output is one and resets both DFF's



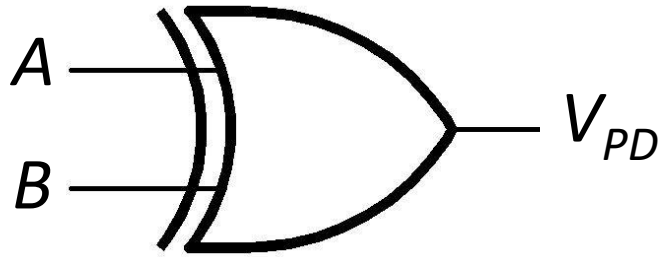
PFD Characteristics



- Phase error characteristic is asymmetric about zero phase, always positive gain
- Output average is positive for $f_A - f_B > 0$, negative for $f_A - f_B < 0$
- Frequency detection feasible



PFD vs. PD

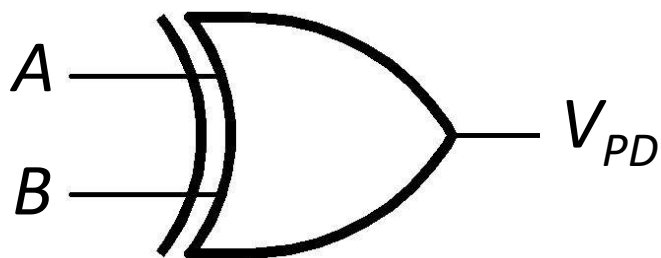


- Simple combinational logic
- ☹️ Level sensitive, duty cycle sensitive
- ☹️ Positive and negative PD gain
- ☹️ V_{PD} output average is zero when input frequency difference is large
- ☹️ Limited acquisition range

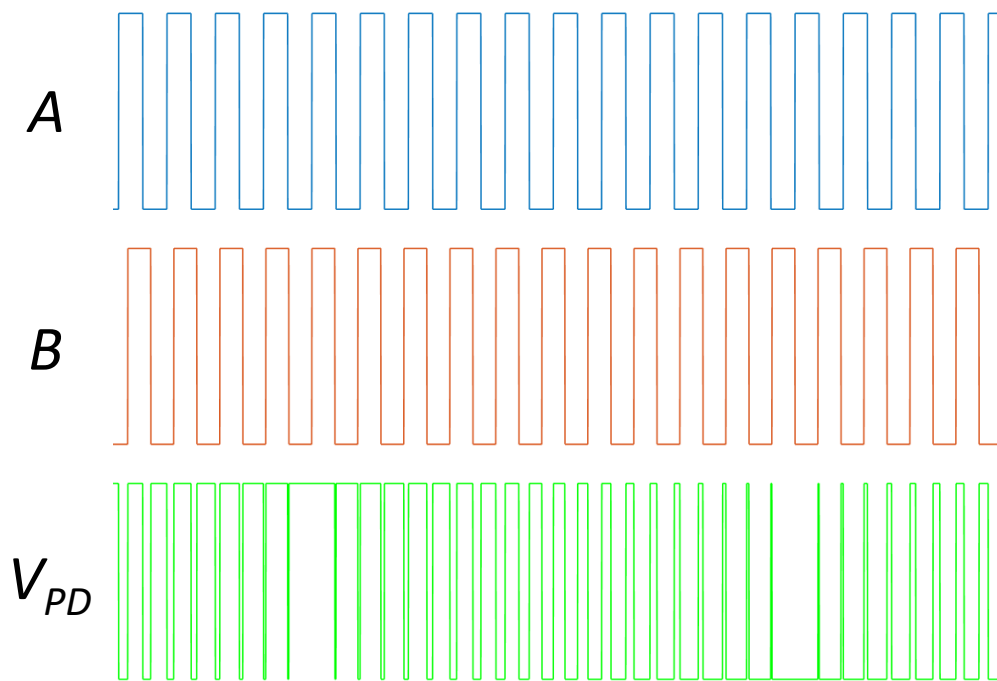
- State machine sequential logic
- 😊 Edge triggered, duty cycle insensitive
- 😊 Positive PD gain only
- 😊 Detect both phase and frequency
- 😊 Unlimited acquisition range



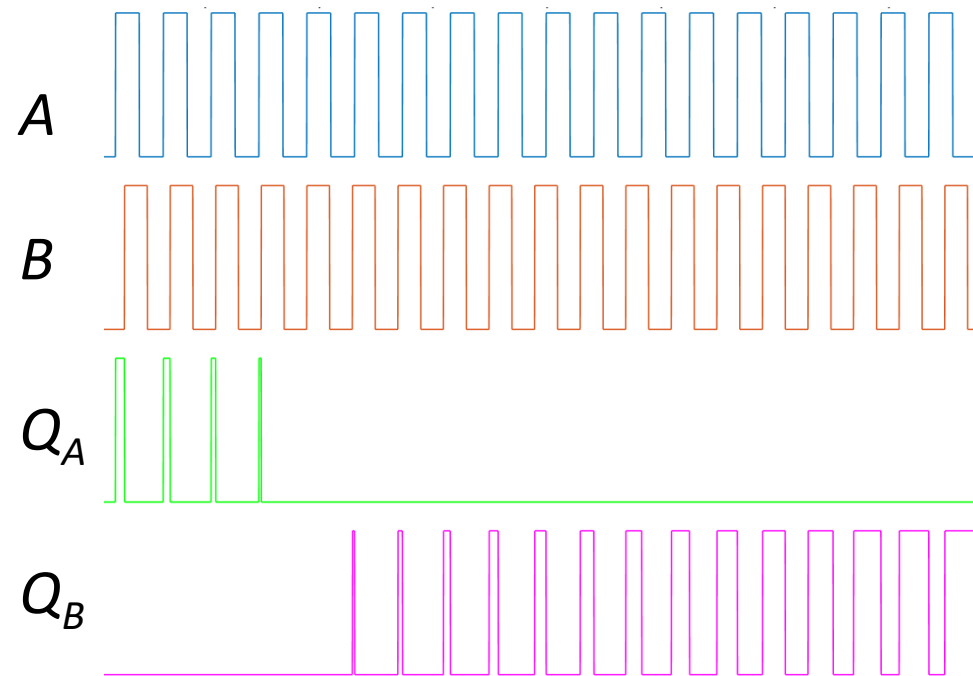
PFD vs. PD with Input Frequency Difference



$$f_A = 0.95 \cdot f_B$$



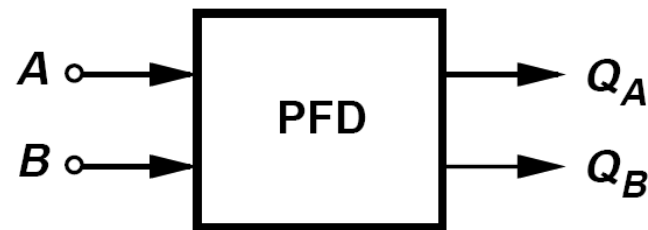
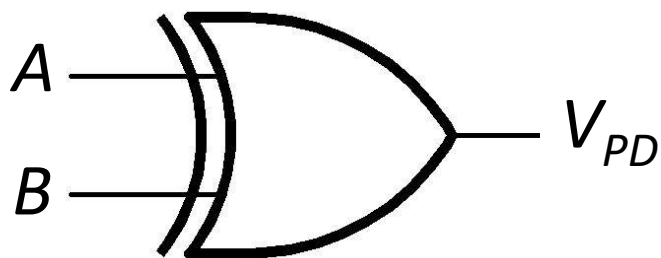
$$\overline{V_{PD}} \approx 0$$



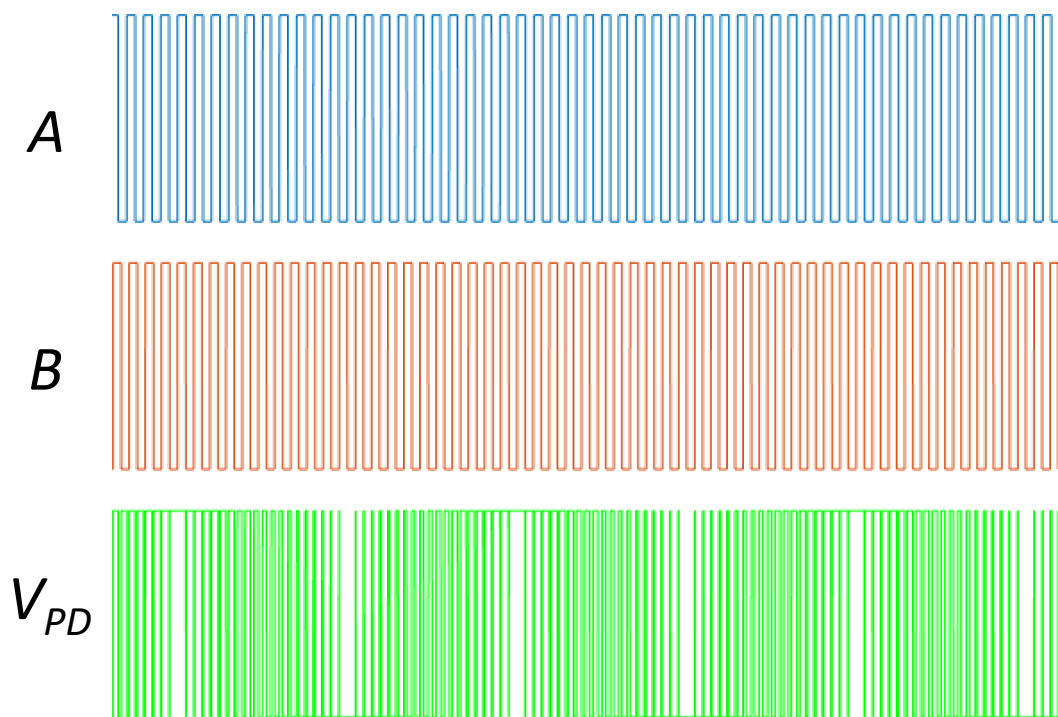
$$\overline{Q_A - Q_B} \propto f_A - f_B$$



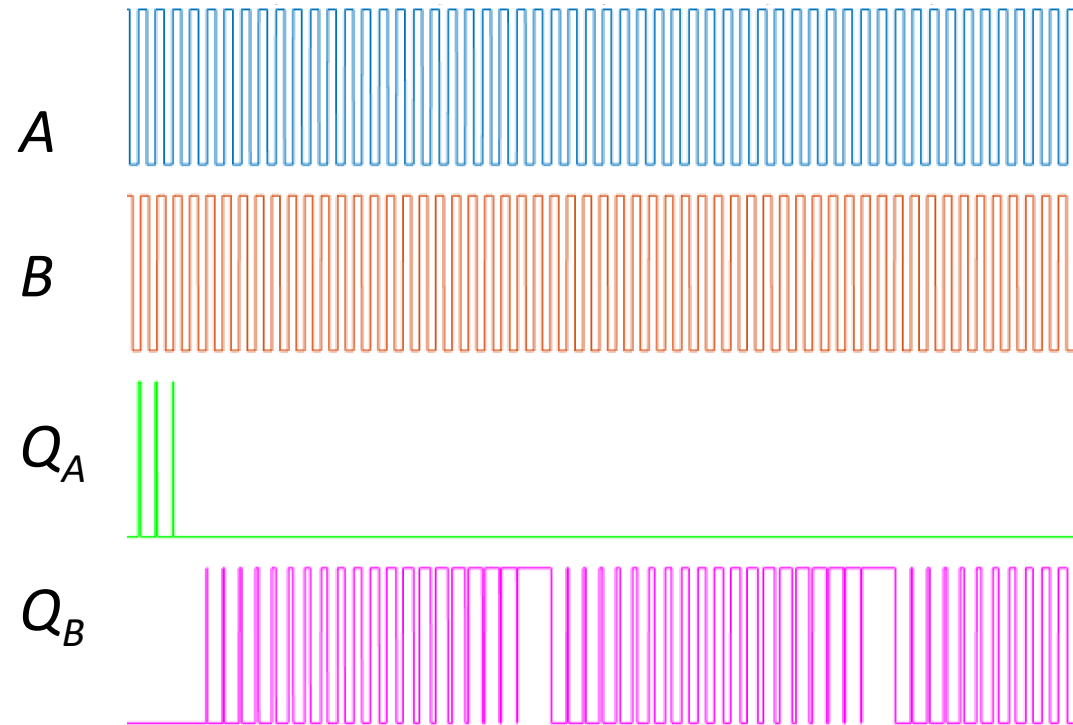
PFD vs. PD with Input Frequency Difference



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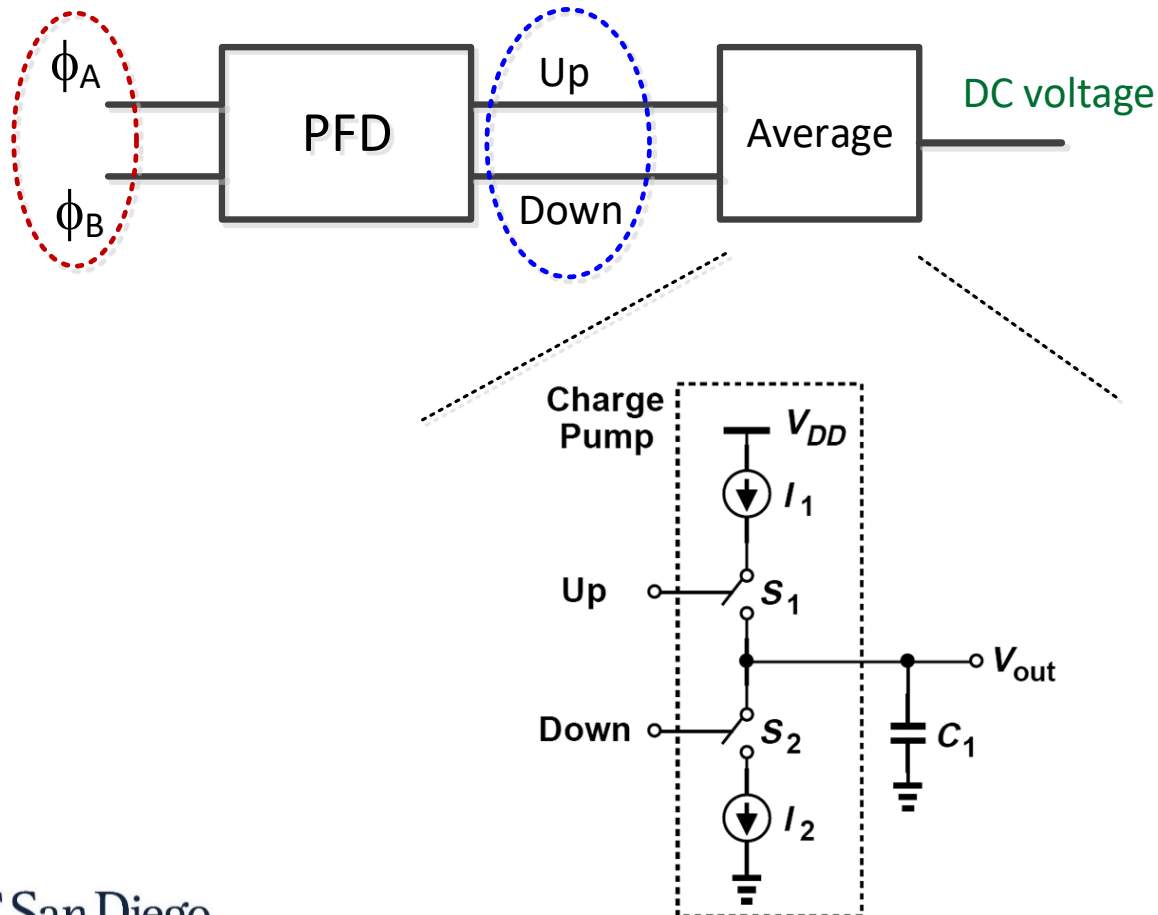
$$\overline{Q_A - Q_B} \propto f_A - f_B$$



Charge Pump

Phase signal
defined by pulse
rising edge

Voltage signal defined
by pulse width



- How to convert the information in pulse width to an average voltage?
 - Use an integrator to average
 - A capacitor charged/discharged by a constant current is an integrator
 - Charge pump current source is turned on by Up and Down signals. Charge time equals to the pulse width

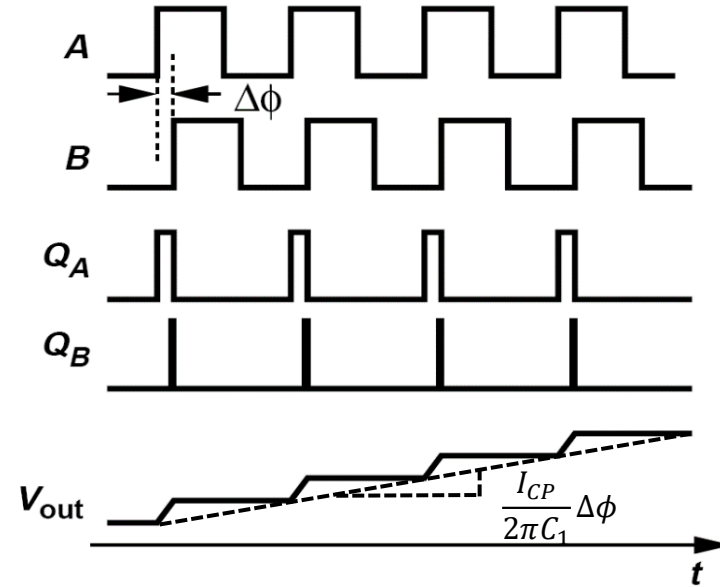
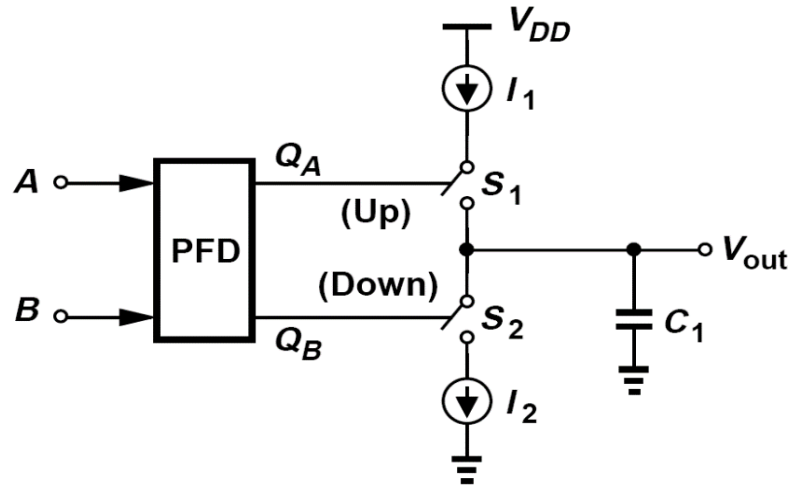
$$\Delta V = \frac{I \cdot \Delta T}{C}$$

$$V_C = \sum_i \frac{I \cdot \Delta T_i}{C}$$

- Pulse width is averaged and converted to the capacitor voltage



Charge Pump



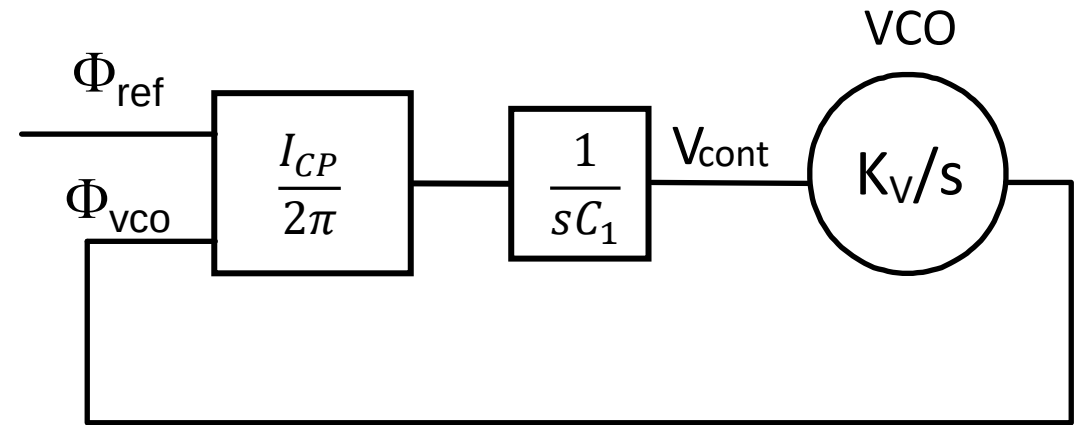
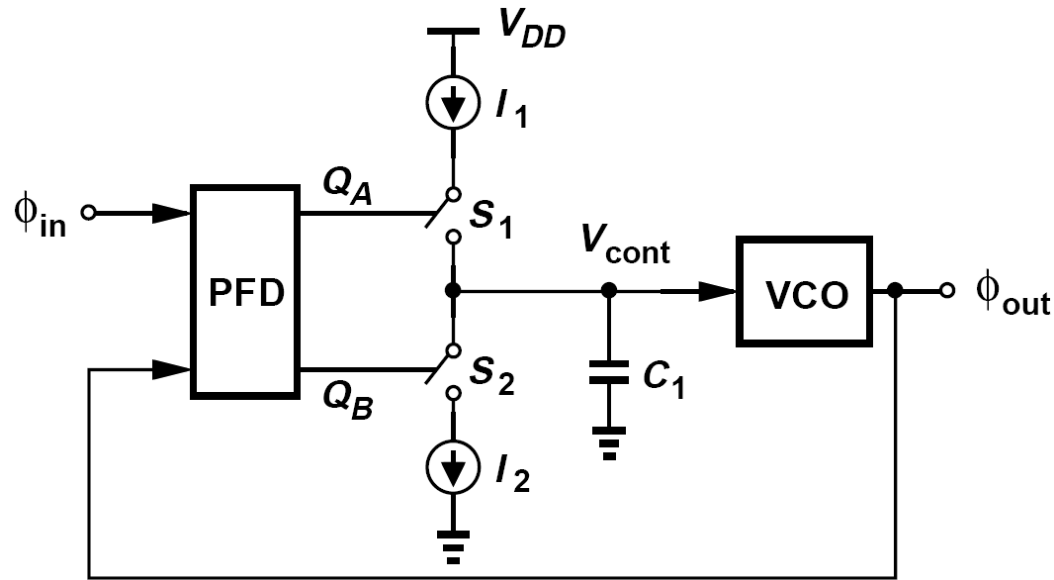
[Razavi]

$$\Delta V_{out} = \frac{I_{CP} \cdot \Delta T}{C_1}, \Delta T = \Delta \phi \frac{T}{2\pi}, \Delta V_{out} = \frac{I_{CP} \cdot T}{2\pi C_1} \Delta \phi$$

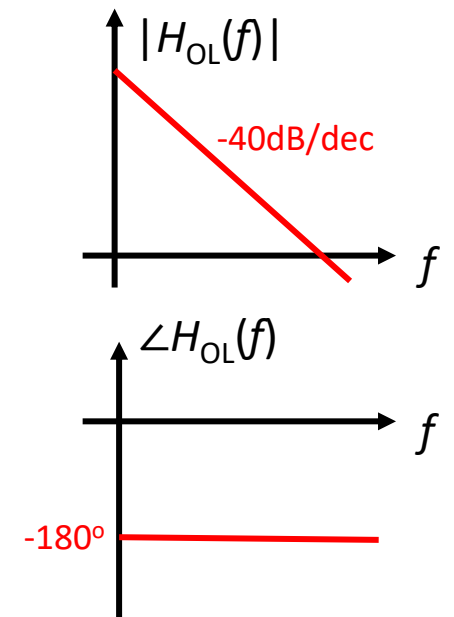
- Approximate V_{out} step waveform by a ramp as if charge pump continuously injects current to C_1 – continuous time approximation
- The slope is approximated as $\frac{\Delta V_{out}}{T} = \frac{I_{CP}}{2\pi C_1} \Delta \phi$
- $V_{out} = \sum_i \frac{I_{CP} \cdot \Delta T_i}{C_1} = \sum_i \frac{I_{CP} \cdot \Delta \phi_i}{2\pi C_1} T$, is approximated by $V_{out} = \int \frac{I_{CP}}{2\pi C_1} \Delta \phi(t) dt \rightarrow \frac{V_{out}}{\Delta \phi}(s) = \frac{I_{CP}}{2\pi C_1 s}$
- Even an arbitrarily small phase $\Delta \phi$ will charge C_1 to saturation over time, indicating an infinite gain of an integrator at DC



Charge Pump PLLs



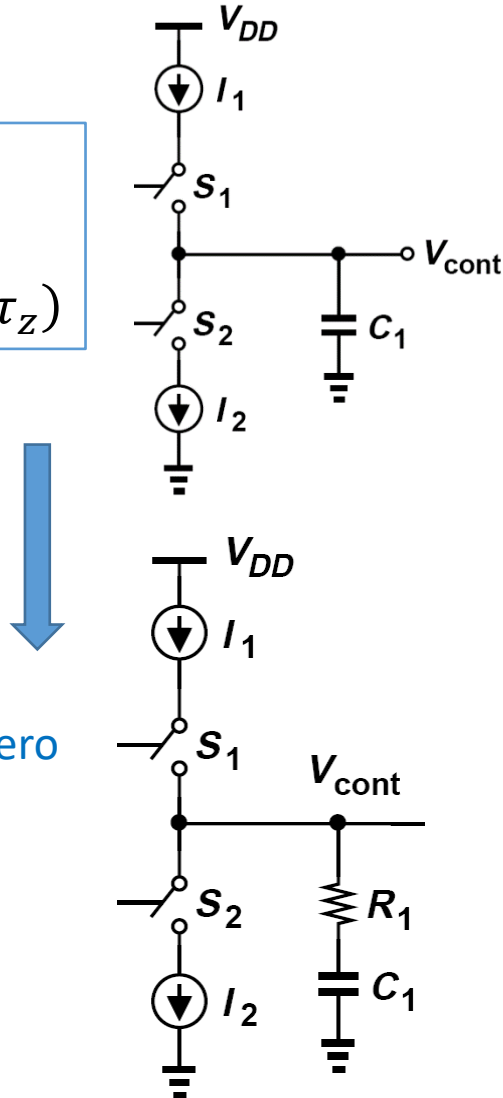
- Integrator by charge pump and $C_1 \rightarrow$ finite $\Delta\phi$ generates infinite $V_{cont} \rightarrow$ no static phase error when loop is locked
- Open loop transfer function $H_{OL}(S) = \frac{I_{CP}K_V}{2\pi C_1 S^2}$
- **Two poles** at DC, one from charge pump + capacitor $\left(\frac{I_{CP}}{2\pi C_1 s}\right)$, one from VCO $\left(\frac{K_V}{s}\right) \rightarrow$ **Type-II PLL**



CPPLL Loop Filter Evolution

A left half plane zero:
 $1 + s\tau_z = 1 + j\omega\tau_z$
 $\angle(1 + j\omega\tau_z) = \arctan(\omega\tau_z)$

- **Adding R1**
- Generates a left half plane zero
- Increases PM
- Stabilizes loop



$$H_{OL}(S) = \frac{I_{CP}K_V}{2\pi C_1 S^2}$$

$$H_{CL}(S) = \frac{H_{OL}}{1 + H_{OL}} = \frac{I_{CP}K_V}{2\pi C_1 S^2 + I_{CP}K_V}$$

- Two poles at DC
- Phase margin = 0
- Loop is not stable

- Adding R_1 in series with C_1 generates a zero

$$H_{OL}(S) = \frac{I_{CP}K_V}{2\pi S} \frac{SR_1 C_1 + 1}{SC_1}$$

$$H_{CL}(S) = \frac{\frac{I_{CP}K_V}{2\pi C_1} (SR_1 C_1 + 1)}{S^2 + \frac{I_{CP}K_V}{2\pi C_1} R_1 S + \frac{I_{CP}K_V}{2\pi C_1}}$$

- Second order system

$$H(s) = s^2 + 2\zeta\omega_n s + \omega_n^2, \omega_n = \sqrt{\frac{I_{CP}K_V}{2\pi C_1}}, \zeta = \frac{R_1}{2} \sqrt{\frac{I_{CP}K_V C_1}{2\pi}}$$



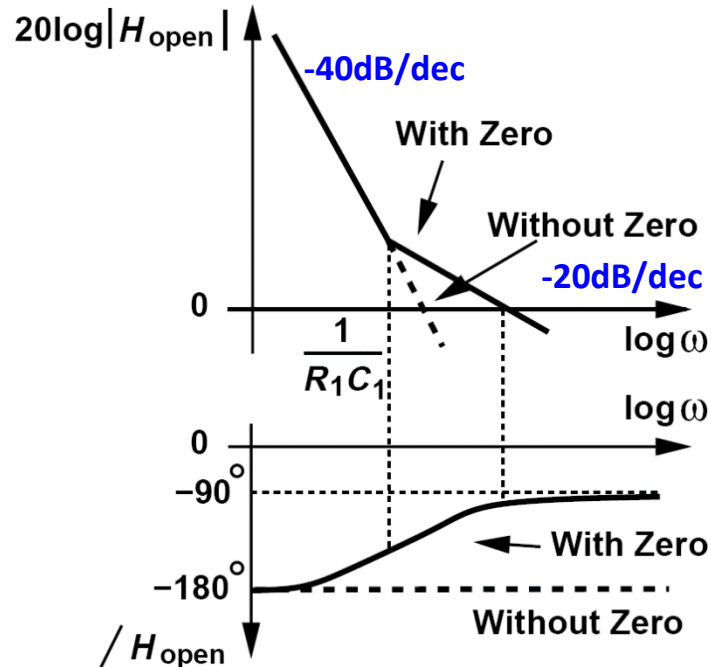
CPPLL Loop Filter Evolution

Type-II PLL

$$H_{OL}(S) = \frac{I_{CP}K_V}{2\pi S} \frac{SR_1C_1 + 1}{SC_1}$$

$$H_{CL}(S) = \frac{\frac{I_{CP}K_V}{2\pi C_1} (SR_1C_1 + 1)}{S^2 + \frac{I_{CP}K_V}{2\pi C_1} R_1S + \frac{I_{CP}K_V}{2\pi C_1}}$$

$$\omega_n = \sqrt{\frac{I_{CP}K_V}{2\pi C_1}}, \zeta = \frac{R_1}{2} \sqrt{\frac{I_{CP}K_VC_1}{2\pi}}$$

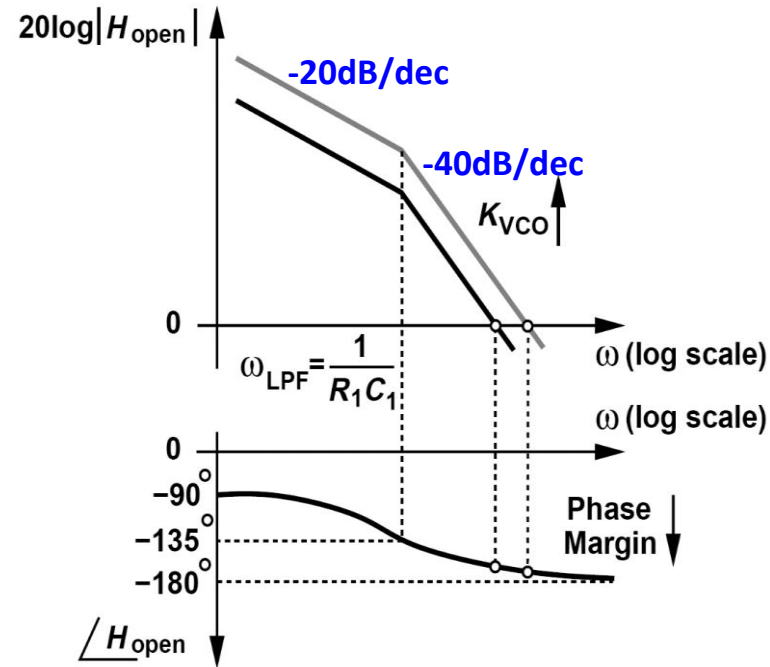


Type-I PLL

$$H_{OL}(s) = \frac{K_{PD} \cdot K_V}{s(1 + sRC)}$$

$$H_{CL}(s) = \frac{\frac{K_{PD} \cdot K_V}{RC}}{s^2 + \frac{s}{RC} + \frac{K_{PD} \cdot K_V}{RC}}$$

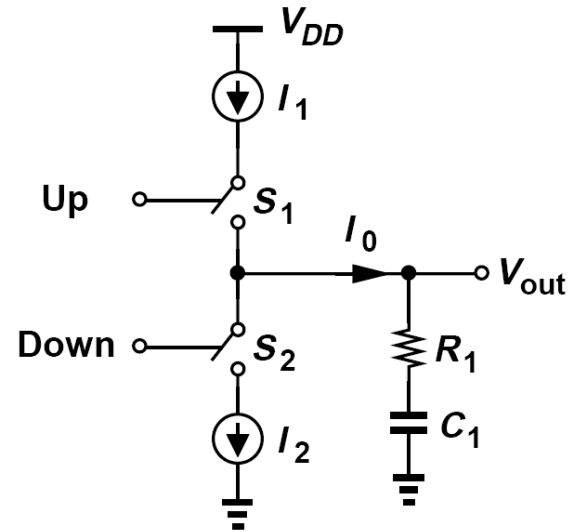
$$\omega_n = \sqrt{\frac{K_{PD} \cdot K_V}{RC}}, \zeta = \frac{1}{2} \sqrt{\frac{1}{K_{PD} \cdot K_V \cdot RC}}$$



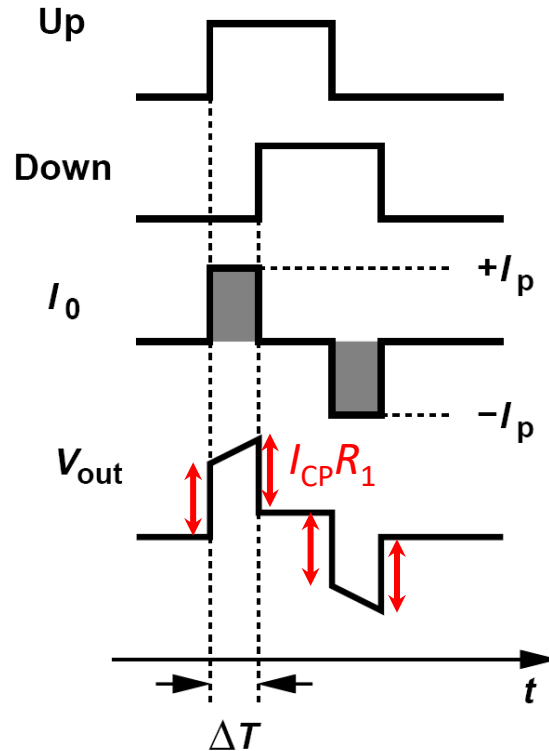
- In Type-II PLL: Real left-half-plane zero — $\frac{1}{R_1C_1}$ increases PM and stabilizes PLL
- Increasing C to reduce BW and ripple
 - Type-I: $\zeta \downarrow$, loop less stable
 - Type-II: $\zeta \uparrow$, loop more stable



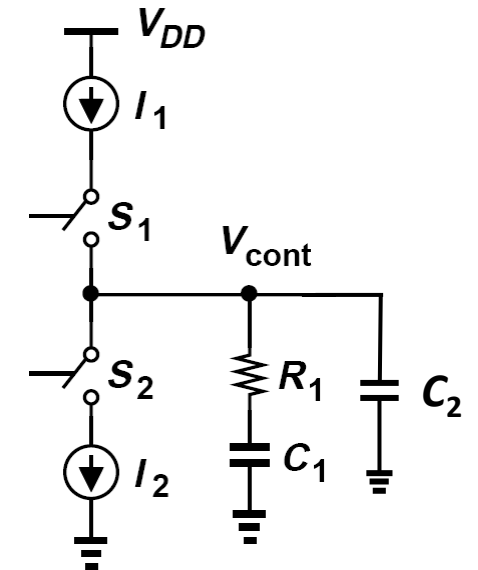
CPPLL Loop Filter Evolution



[Razavi]



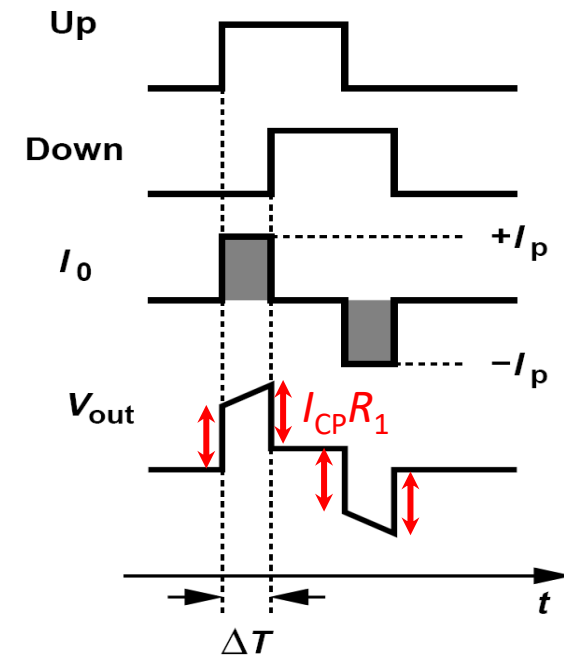
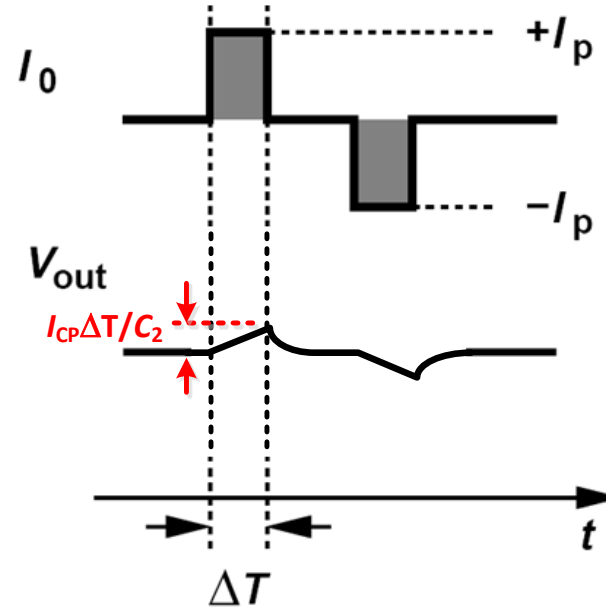
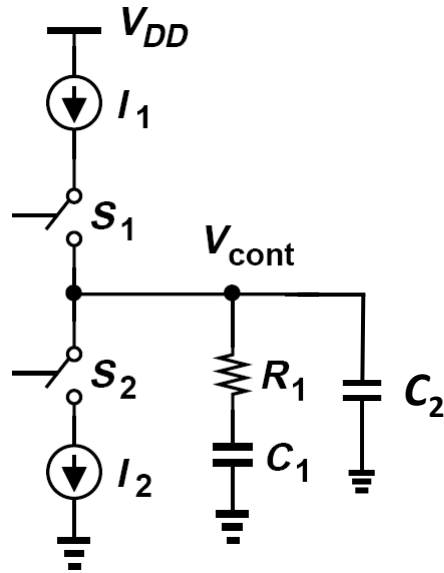
- Adding C_2
- Low impedance path for CP current
- Reduces V_{out} ripple
- Degrades PM



- ☺ Adding R_1 in series with C_1 generates a zero and stabilizes loop
- ☹ When Up and Down signals have timing mismatch, charge pump output current pulses generate voltage jump or drop of $I_{CP}R_1$ in V_{out} ripple and spur!



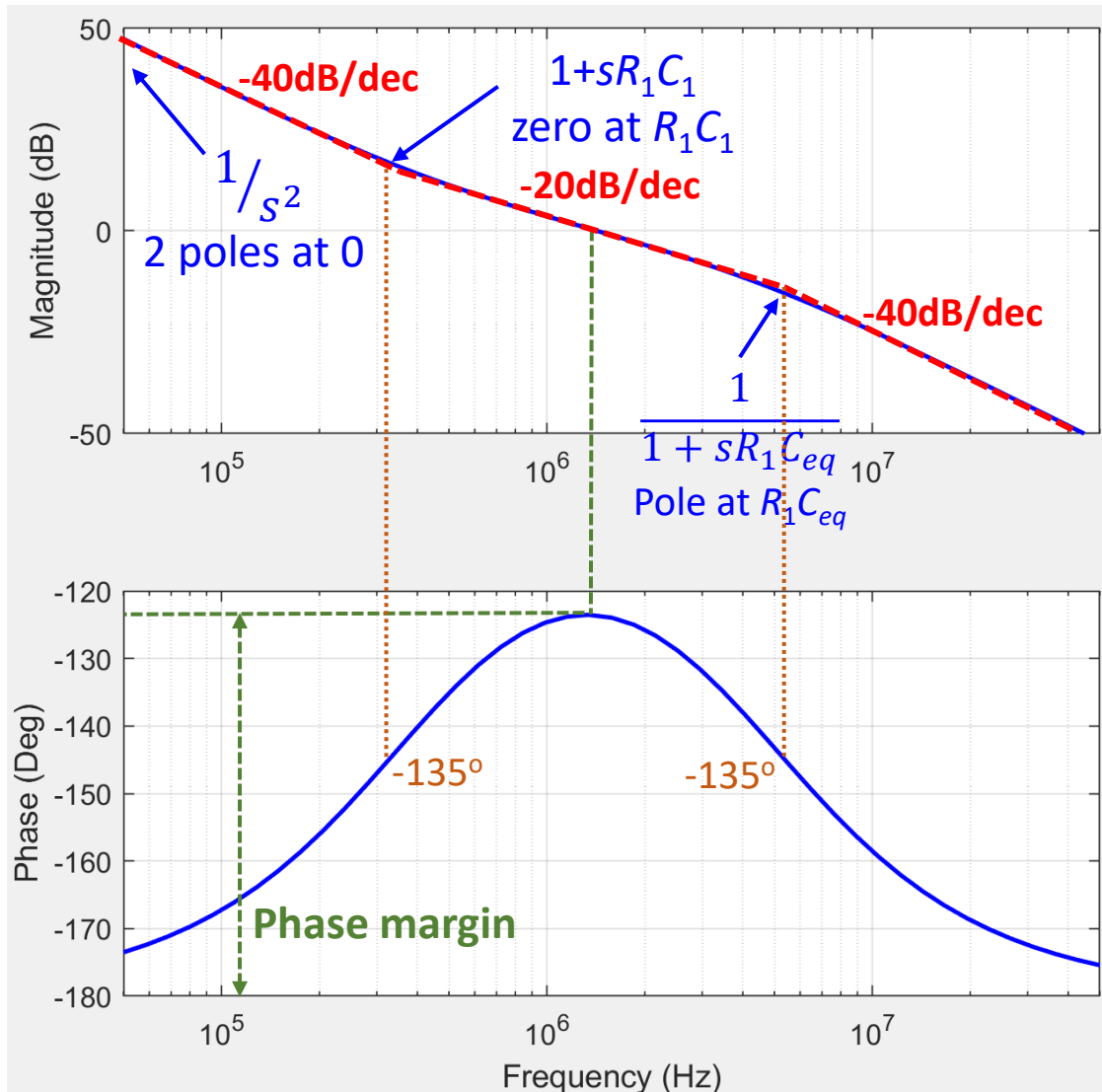
CPPLL Loop Filter Evolution



- Adding C_2 reduces V_{cont} ripple.
 - A low impedance path for charge pump current pulses ($\frac{1}{\omega_{\text{ref}} C_2} \ll R_1$)
 - Ripple is reduced from $I_{\text{cp}} R_1$ to $I_{\text{cp}} \Delta T / C_2$.
- C_2 generates an additional pole at $R_1 \frac{C_1 C_2}{C_1 + C_2}$ that degrades phase margin.
 - Manageable with proper C_1 and C_2 ratio.



CPPLL Loop Stability



- Open loop transfer function

$$H_{OL}(S) = \frac{I_{CP}K_V}{2\pi s} \frac{1}{s(C_1 + C_2)} \frac{1 + SR_1C_1}{1 + SR_1 \frac{C_1C_2}{C_1 + C_2}}$$

$$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$$

$$\tau_z = R_1C_1, \tau_p = R_1C_{eq} \approx R_1C_2$$

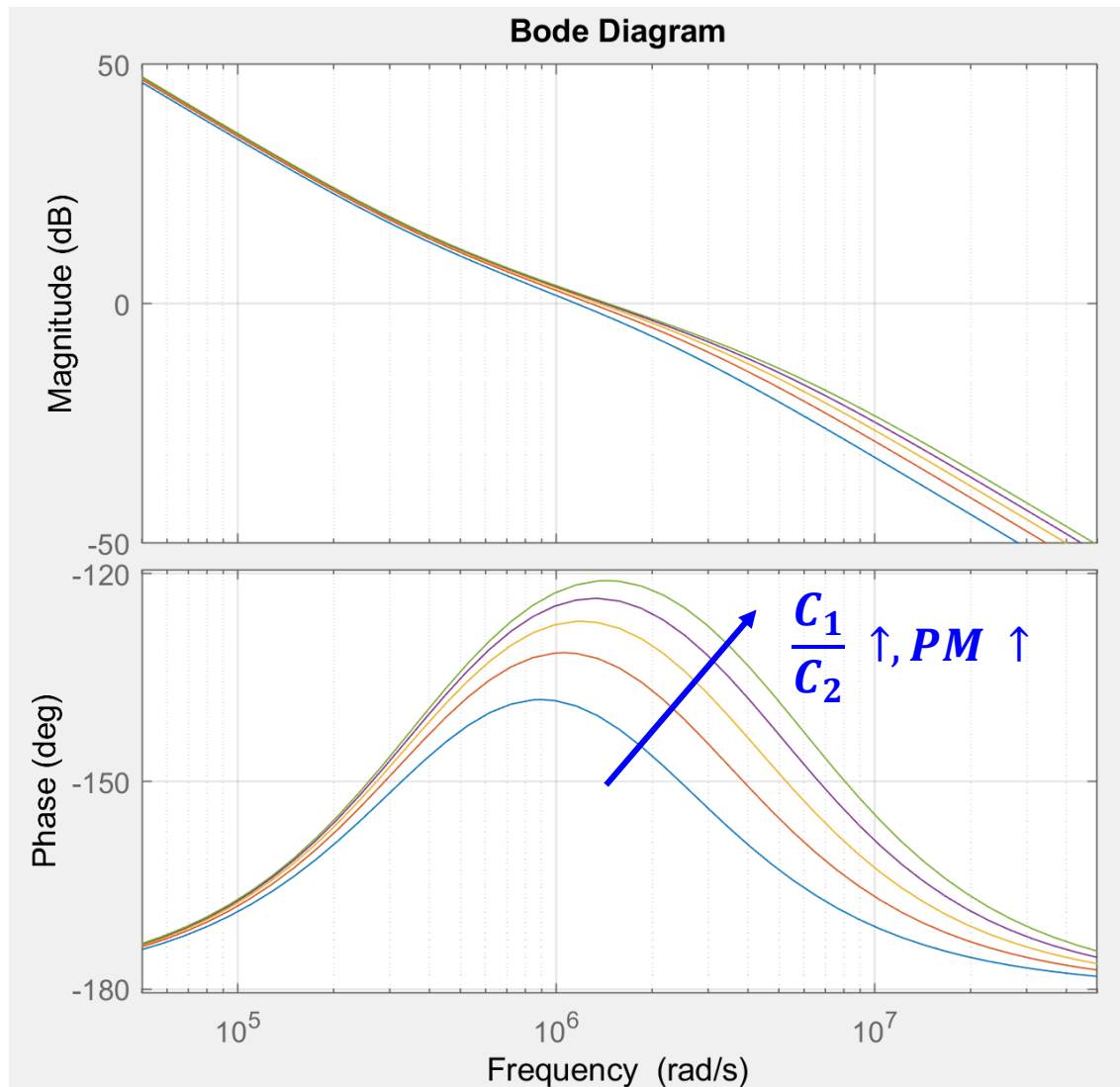
- To obtain enough phase margin:

$$\tau_p \ll \tau_z \rightarrow C_1 \gg C_2$$

- $PM_{max} = \tan^{-1} \left(\frac{C_1/C_2}{2\sqrt{1+C_1/C_2}} \right)$



CPPLL Loop Stability

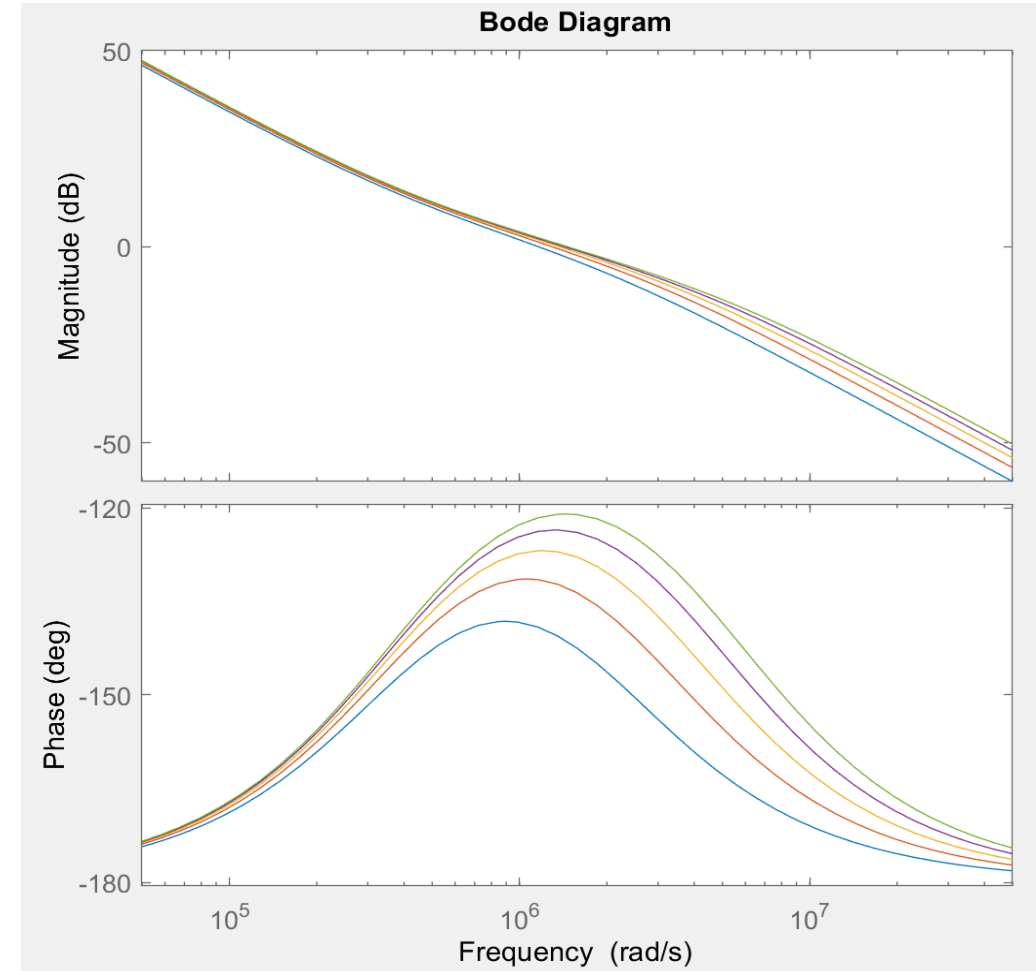


- To obtain enough phase margin:
 $\tau_p \gg \tau_z \rightarrow C_1 \gg C_2$
- $PM_{max} = \tan^{-1} \left(\frac{C_1/C_2}{2\sqrt{1+C_1/C_2}} \right)$
- Larger C_1/C_2 ratio yields larger phase margin at the cost of larger capacitor size
- Loop bandwidth and phase margin optimization is decoupled



Type-II PLL S-Domain Model Open Loop TF

```
% Loop parameters
lcp=20e-5;
kv0=1e6;
c2=50e-12;
for k=4:2:12
    kv=kv0*k;
    c1=c2*k;
    r1=5e4/k;
    tz=r1*c1;
    tp=r1*(c1*c2)/(c1+c2);
% Transfer function
    num=[tz*lcp*kv lcp*kv];
    den=[2*pi*(c1+c2)*tp 2*pi*(c1+c2) 0 0];
    H=tf(num, den);
% Bode plot
    bode(H, {5e4, 5e7}); hold on;
    [gm, pm, wgm, wpm] = margin(H);
    pm1(k) = pm;
end
```



Summary

- Type-I PLL $\rightarrow$ Type-II PLL
- Three issues in Type-I PLL solved
 - Finite locking range: PFD can detect both phase and frequency difference with only positive PFD gain
 - Static phase error in the steady state: introduce a second pole at DC $\rightarrow$ Type-II PLL
 - Loop stability (damping factor or PM) is coupled to loop gain and loop bandwidth: R1, C1 and C2 loop filter configuration provides a zero and pole pair that makes PM only determined by C1/C2 ratio



References

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- William F. Egan, Frequency Synthesis by Phase Lock, 2nd edition, 2000
- Michael Perrott, MIT Opencourseware, High Speed Communication Circuits
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